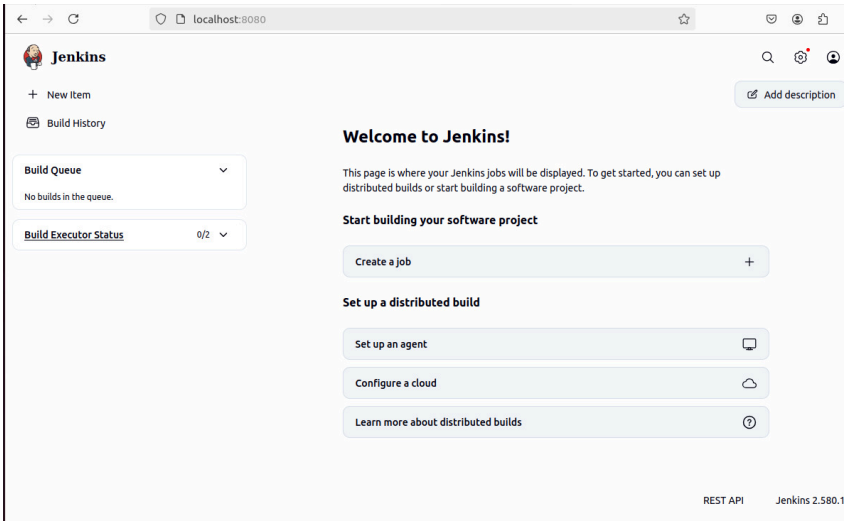
Vidyalankar Institute of Technology Accredited A+ by NAAC	Department of Computer Engineering A.Y-2026-27

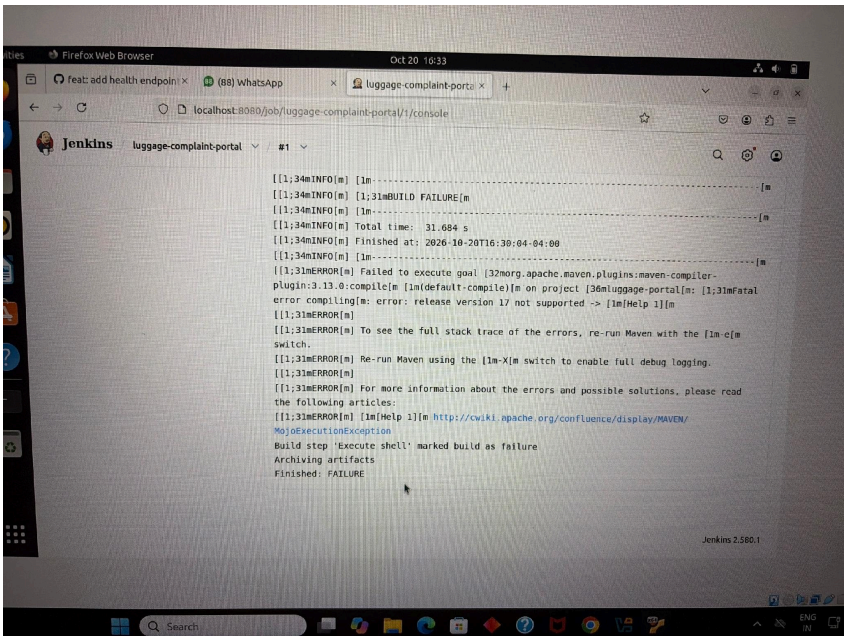
Semester	B.E. Semester VII – CMPN
Subject	DevOps
Subject Professor In-charge	Prof. Snehal Andhare
Assisting Teachers	Prof. Snehal Andhare
Laboratory	

Student Name	Vaishnavi Khopkar	
Roll Number	23102A0074	
Grade and Subject Teacher's Signature		

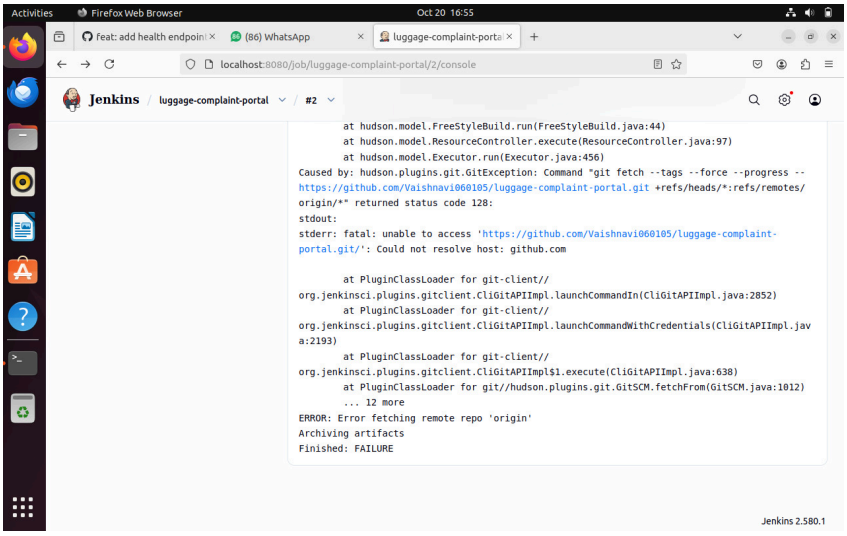
Experiment Number	7
Experiment Title	Luggage Complaint Portal: Jenkins Installation and Continuous Integration Job
Objectives (Skill Set / Knowledge Tested / Imparted)	To install and configure Jenkins, connect the GitHub repository, create a Maven build job triggered by commit polling, and archive the build artefact.
Program	**1. Jenkins installed**  *Figure: Jenkins dashboard after installation (Jenkins 2.580.1)* **2. Job configuration** • Job type: Freestyle project (luggage-complaint-portal). • Source: https://github.com/Vaishnavi060105/luggage-complaint-portal.git, branch */develop. • Trigger: Poll SCM with schedule H/2 * * * *. • Build step: Execute shell using Java 17, then mvn clean package -DskipTests. • Post-build: Archive the artifacts target/*.war.

```
export JAVA_HOME=/usr/lib/jvm/java-17-openjdk-amd64
export PATH=$JAVA_HOME/bin:$PATH
mvn clean package -DskipTests
```

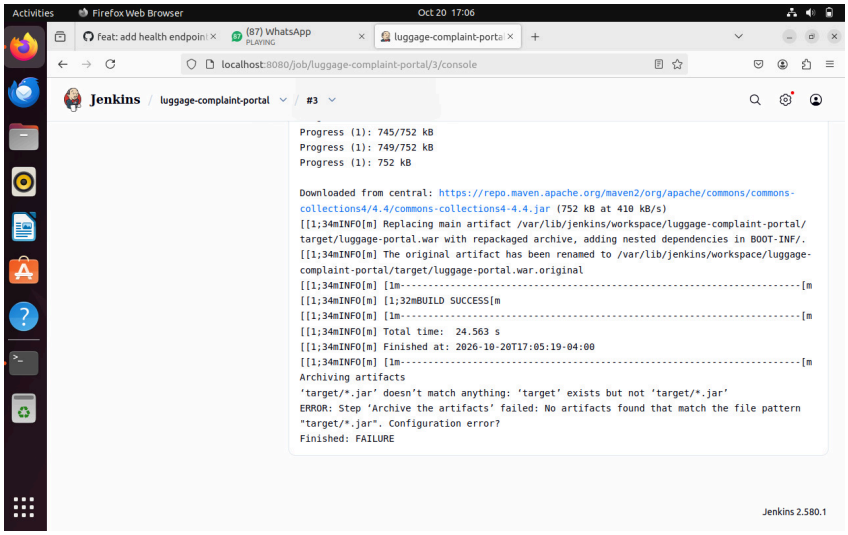
3. Failed builds and their fixes (troubleshooting)



Build #1: release version 17 not supported (default Java had no compiler); fixed by setting JAVA_HOME to Java 17



Build #2: Could not resolve host github.com (VM DNS); fixed by setting a public DNS



Build #3: BUILD SUCCESS but target/.jar not found; the project builds a WAR, so the pattern was changed to target/*.war*

4. Successful build and archived artefact

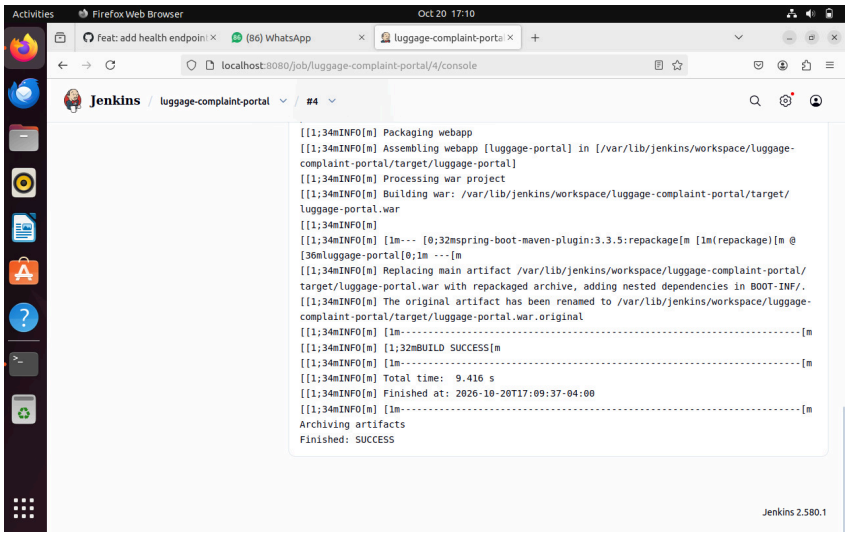


Figure: Console output of build #4 ending in BUILD SUCCESS

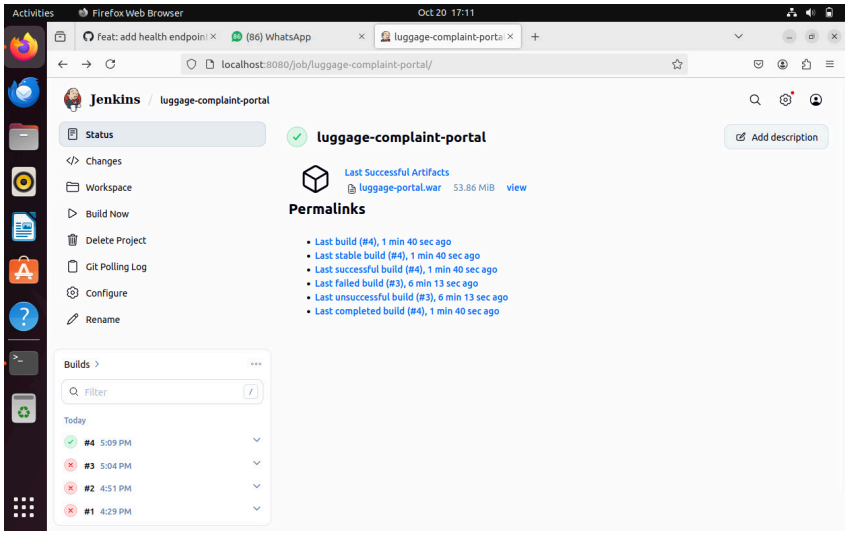


Figure: Job page with builds #1 to #4 and the archived luggage-portal.war

5. Trigger evidence

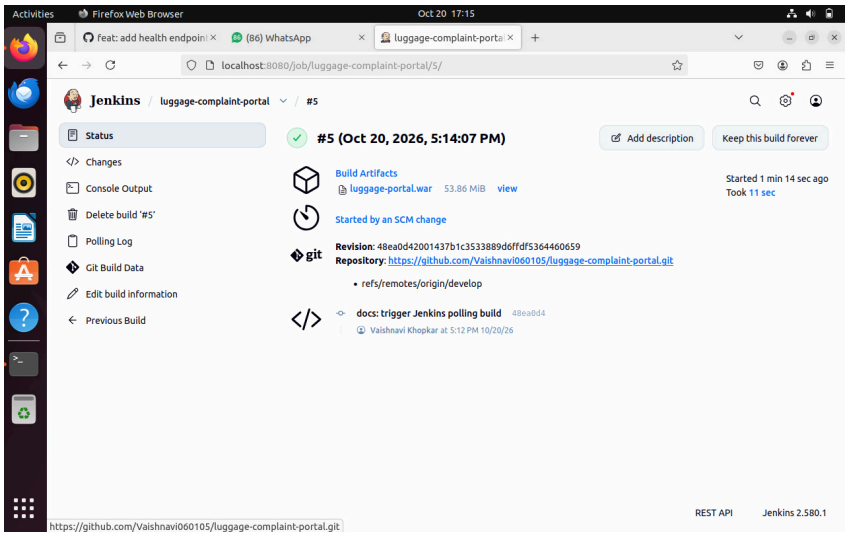


Figure: Build #5 started by an SCM change after a commit to develop

Conclusion	Jenkins was installed and a Maven CI job was connected to GitHub, started automatically by commits through SCM polling, and archived the WAR artefact. The failed builds were diagnosed and fixed.